Northstar Knowledge Assistant — Use cases

Appendix to the project deck · Version 1.0 · 2026-11-04-17-00-00 UTC
Component IDs refer to the catalog (project deck, §7). ★ = riskiest use case, designed and prototyped first.

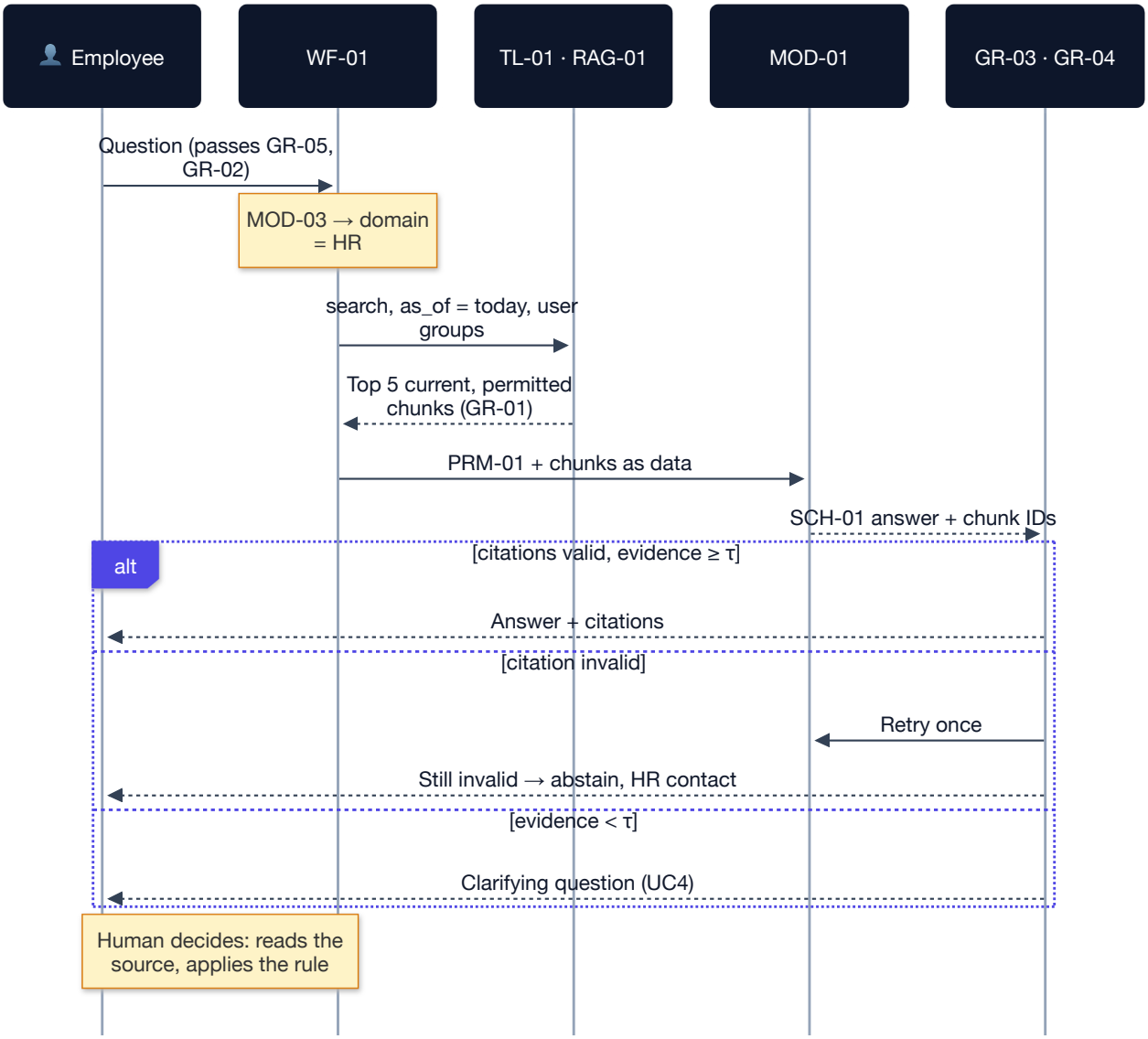
#	Use case	Pattern	Autonomy	Priority
UC1	Policy Q&A, version-aware	Workflow `WF-01`	Suggests; the employee decides	★ riskiest
UC2	Product troubleshooting lookup	Workflow `WF-01`	Suggests; the analyst decides	High
UC3	Policy change comparison	Workflow `WF-02`	Suggests; the content owner validates	Medium
UC4	Low-evidence abstention and gap report	Workflow `WF-01` (abstain branch)	Acts alone (logs the gap)	High

UC1 ★ Policy Q&A, version-aware

1. Story

> "As an **employee**, when **I need to know whether an expense or a leave rule applies to me**, I want **a short answer quoting the current policy clause**, so that **I act correctly without opening five PDFs or waiting for HR**."

2. User flow



Step	Human	AI	Tool / system	Autonomy
1	Asks the question	—	`GR-05`, `GR-02`	—
2	—	Classifies the domain (`MOD-03`)	—	Acts alone (internal)
3	—	—	`TL-01` with `GR-01` filters	Code
4	—	Drafts a cited answer (`MOD-01`)	—	Suggests
5	—	—	`GR-03`, `GR-04` validate	Code
6	**Reads the sources and decides**	—	Citation opens `TL-02` excerpt	Human

On failure: retrieval timeout → retry once, then "service unavailable, contact HR"; validator fails twice → abstain; conflicting current sources → show both and route to the content owner (`POL-01`).

3. Context

Step	Need	Component	Freshness & access
3	Policy chunks	`RAG-01` over `DOC-01`, built by `WF-03`	Re-indexed on publish (≤ 1 h); filtered by `access_groups` and `as_of`
4	Instructions	`PRM-01`, `POL-01`	Versioned in git; version logged in `OB-01`
4	Conversation state	`MEM-01`	Session only; **no** personal data stored
5	Output contract	`SCH-01`	Validated by code

Context budget: top 5 chunks (≈ 2.5k tokens). Relevance beats volume.

4. Tools

Tool	Purpose	Read / act
`TL-01` `search_knowledge`	Filtered hybrid retrieval	Read
`TL-02` `fetch_document_section`	Exact excerpt behind each citation	Read

The model only *requests* a search. Code fills `as_of` and `groups` from the session, never from the model, and rejects `top_k` > 10. No act tools: the assistant never changes a policy or files a claim.

5. Flow and models

Workflow (`WF-01`): the steps are known in advance, so code decides them. An agent (`AG-01`) is tested only in `XP-02`.

AI step	Model	Why this model
Domain routing	`MOD-03` classifier	Narrow 4-class task; cheap and measurable (accuracy ≥ 95 %)
Retrieval	`MOD-02` embeddings + `MOD-04` reranker	Semantic match plus precise ordering
Answer	`MOD-01` LLM	Language synthesis with citations

6. Guardrails

Input	Execution	Output
`GR-05` blocks personal-record requests ("show Ana's salary"); `GR-02` flags injected text in queries and chunks	`GR-01` filters by access group and effective date **in code**, before the model sees anything	`GR-03` every claim cites a retrieved chunk; `GR-04` abstains below τ; safe fallback = HR contact

False alarms are tested too: `GR-05` must not block "how many leave days do I have under the policy?".

7. Evals

Test set: 40 gold questions: 20 normal, 8 hard (multi-document), 6 edge (version-sensitive), 6 adversarial (unanswerable or injected).

Dimension	Metric	Evaluator	Threshold
Retrieval	recall@5 / MRR	`EV-01` code	≥ 85 % / ≥ 0.7
Correctness	Answer matches the reference rubric	`EV-03` judge (`MOD-05`), calibrated on 10 human labels	≥ 3.5 / 4

Dimension	Metric	Evaluator	Threshold
Groundedness	Citation IDs resolve and support the claim	`EV-02` code + judge	100 % resolve; ≥ 90 % support
Policy	Superseded or unauthorised source used	`EV-04` code	0

8. Observability

Signal	What is logged	Alert / review trigger
Traces	Query (pseudonymised), filters, chunk IDs + versions, prompt and model version (`OB-01`)	Any answer citing an expired version
Speed, errors, cost	p95 latency, retries, tokens, € per answer	p95 > 6 s; cost > €0.05 per answer
Quality	Judge score on a 2 % sample, 👍 /👎 (`OB-02`)	Weekly judge mean < 3.5; 👎 rate > 10 %

Every 👎 with a reason becomes a gold-set candidate.

9. Experiments

Hypothesis	Variant vs. baseline	Dataset	Decision metric
Hybrid search finds policy codes that vectors miss	`XP-01` hybrid vs. vector-only	40 gold questions	recall@5 +5 pts
An agent finds multi-doc answers better	`XP-02` `AG-01` vs. `WF-01`	8 hard questions	Groundedness at ≤ 1.5× cost

10. Risks and open questions

- Two "current" versions overlap in dates → both shown; does `POL-01` cover every case? (Owner: Head of People Ops.)
- Users may stop reading citations → measure the citation-click rate in the pilot.
- Is τ (abstention threshold) too strict? Tune it on the gold set; watch the abstention rate in `OB-02`.

UC2 Product troubleshooting lookup

1. Story: "As a **support analyst**, when **a customer reports an NX-200 that will not boot**, I want **the reset steps from the right manual version**, so that **I fix it on the first call**."

2. User flow

Step	Human	AI	Tool / system	Autonomy
1	Asks; picks the product if prompted	`MOD-03` → domain = product	—	Suggests
2	—	—	`TL-01` filtered by `product_id` + `GR-01`	Code
3	—	`MOD-01` writes numbered steps with citations	`GR-03`	Suggests
4	**Follows or rejects the steps**	—	`TL-02` opens the manual page	Human

Product ambiguous ("my NX") → clarifying question, not a guess.

3. Context: `RAG-01` over the 12 manuals in `DOC-01` (`product_ids` metadata; re-indexed by `WF-03` on publish); `MEM-01` keeps `product_id` for follow-ups; output `SCH-01`.

4. Tools: `TL-01`, `TL-02` (read). No ticket tool here: that is Example 2.

5. Flow and models: `WF-01`; `MOD-02` + `MOD-04` for ranking (procedure steps are lexically close across models); `MOD-01` formats the steps.

6. Guardrails: `GR-02` on manual text; `GR-03` citations; product mismatch between query and chunk → block.

7. Evals: 12 gold questions in `EV-01`; wrong-product citation = 0 (code); step completeness ≥ 3.5/4 (`EV-03`).

8. Observability: `OB-01` product ID per run; alert when the same product gets 👎 three times in a week.

9. Experiments: `XP-03` reranker on/off; decision = MRR gain ≥ 0.05 with p95 latency still within 6 s.

10. Risks: manuals for old firmware are still "current" for old units → is `as_of` the right filter, or should it be the firmware version? (Open.)

UC3 Policy change comparison

1. Story: "As a **content owner**, when **a new expense policy version is published**, I want **a clause-by-clause change report**, so that **I can tell employees exactly what changed**."

2. User flow

Step	Human	AI	Tool / system	Autonomy
1	Picks the policy and two versions	—	—	—
2	—	—	`TL-03` deterministic diff	Code
3	—	`MOD-01` summarises the impact of each change → `SCH-02`	`GR-03`	Suggests
4	**Validates and publishes the notice**	—	—	Human

3. Context: both versions of the `DOC-01` policy via `TL-02`; `POL-01` decides which is authoritative; output `SCH-02`.

4. Tools: `TL-02`, `TL-03` (read). **5. Flow:** `WF-02`. The diff is code (exact); the LLM only explains it, because a model diff would miss clauses.

6. Guardrails: every change item must map to a diff hunk (`GR-03`); no hunk left unexplained.

7. Evals: 3 seeded policy pairs; diff recall 100 % (code); summary faithfulness ≥ 3.5/4 (`EV-03`); `EV-04` version fixtures.

8. Observability: `OB-01` logs the versions compared. **9. Experiments:** none for V1: the diff is deterministic and there is one design.

10. Risks: reformatted but unchanged clauses show up as noise → normalise whitespace before the diff.

UC4 Low-evidence abstention and gap report

1. Story: "As an **employee**, when **the policies do not cover my question**, I want **to be told so, with a next step**, so that **I do not act on an invented rule**. As a **content owner**, I want **those gaps listed**."

2. User flow

Step	Human	AI	Tool / system	Autonomy
1	Asks	`MOD-03` routes	`TL-01`	—
2	—	—	`GR-04`: evidence score < τ	Code
3	—	`MOD-01` writes a clarifying question	`SCH-01` `status=insufficient`	Suggests
4	Clarifies or contacts HR	—	Gap logged to `OB-02`	Acts alone (logging)
5	**Content owner triages the gap list weekly**	—	—	Human

3. Context: `RAG-01` over `DOC-01`, `SCH-01`, `PRM-01` (the abstention instruction). **4. Tools:** `TL-01` (read). Logging is internal, not a tool.

5. Flow: `WF-01` abstain branch. The decision to abstain is a **threshold in code**, not the model's choice.

6. Guardrails: `GR-04` threshold; `GR-05` scope; output never contains a policy claim without a citation.

7. Evals: 6 unanswerable + 4 near-miss questions; correct abstention ≥ 90 %; false abstention on answerable questions ≤ 5 % (`EV-03`, `EV-01`).

8. Observability: abstention rate per domain; a spike means a missing document or a broken index.

9. Experiments: τ sweep (0.3–0.6) on the gold set; decision = best balance of correct and false abstention.

10. Risks: too many abstentions → users give up. The pilot target is an abstention rate < 15 % on real traffic.